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EEE 554 — Lecture #9

Beta Distribution

For $a > 0$ and $b > 0$, the beta function is defined by:

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

A random variable (RV) X is beta distributed with parameters $a > 0$ and $b > 0$ if:

$$f_X(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$$

for $0 \leq x \leq 1$ and zero otherwise. This is expressed by the notation $X \sim \beta(a, b)$.

Note that, for $X \sim \beta(a, b)$:

$$\int_{-\infty}^{\infty} f_X(x) dx = \frac{1}{B(a, b)} \int_0^1 x^{a-1} (1-x)^{b-1} dx = \frac{B(a, b)}{B(a, b)} = 1$$

and

$$\begin{aligned} E[X] &= \frac{1}{B(a, b)} \int_0^1 x^a (1-x)^{b-1} dx = \frac{B(a+1, b)}{B(a, b)} \\ &= \frac{\Gamma(a+1)\Gamma(b)/\Gamma(a+b+1)}{\Gamma(a)\Gamma(b)/\Gamma(a+b)} = \frac{a}{a+b} \end{aligned}$$

Covariance of Two RVs

The covariance of X_1 and X_2 is:

$$\text{Cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]$$

where $\mu_1 = E[X_1]$ and $\mu_2 = E[X_2]$.

Note that:

$$\begin{aligned} \text{Cov}(X_1, X_2) &= E[X_1 X_2 - X_1 \mu_2 - \mu_1 X_2 + \mu_1 \mu_2] \\ &= E[X_1 X_2] - \mu_1 \mu_2 \end{aligned}$$

RVs X_1 and X_2 are uncorrelated if $\text{Cov}(X_1, X_2) = 0$; i.e., if $E[X_1 X_2] = \mu_1 \mu_2$.

Theorem: Independent RVs are uncorrelated.

Proof:

$$\begin{aligned}
E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x,y) dx dy && \text{(Expectation)} \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) f_Y(y) dx dy && \text{(Independence)} \\
&= \int_{-\infty}^{\infty} x f_X(x) dx \int_{-\infty}^{\infty} y f_Y(y) dy \\
&= E[X]E[Y]
\end{aligned}$$

PDF of the Sum of Independent RVs

Suppose X_1 and X_2 are independent and $Z = X_1 + X_2$. Then:

$$\begin{aligned}
F_Z(z) &= P(\{Z \leq z\}) = P(\{X_1 + X_2 \leq z\}) \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x_1} f_{X_1,X_2}(x_1, x_2) dx_2 dx_1
\end{aligned}$$

Since X_1 and X_2 are independent:

$$F_Z(z) = \int_{-\infty}^{\infty} f_{X_1}(x_1) \left[\int_{-\infty}^{z-x_1} f_{X_2}(x_2) dx_2 \right] dx_1$$

Using the fundamental theorem of calculus:

$$\begin{aligned}
f_Z(z) &= \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} f_{X_1}(x_1) \left[\frac{d}{dz} \int_{-\infty}^{z-x_1} f_{X_2}(x_2) dx_2 \right] dx_1 \\
&= \int_{-\infty}^{\infty} f_{X_1}(x_1) f_{X_2}(z-x_1) dx_1 \\
&= [f_{X_1} * f_{X_2}](z)
\end{aligned}$$

i.e., the convolution of f_{X_1} and f_{X_2} .

Example

Let the joint PDF be:

$$f_{X_1,X_2}(x_1, x_2) = \begin{cases} 3/4 & 0 \leq x_1 \leq 1, 0 \leq x_2 \leq x_1^2 \\ 3/4 & 1 \leq x_1 \leq 2, 0 \leq x_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Marginal of X_2 :

$$f_{X_2}(x_2) = \int_{\sqrt{x_2}}^2 \frac{3}{4} dx_1 = \frac{3}{4}(2 - x_2^{1/2}), \quad 0 \leq x_2 \leq 1$$

Expected value of X_2 :

$$E[X_2] = \frac{3}{4} \int_0^1 x_2(2 - x_2^{1/2}) dx_2 = \frac{3}{4} \int_0^1 (2x_2 - x_2^{3/2}) dx_2 = \frac{9}{20}$$

2. Marginal of X_1 :

$$f_{X_1}(x_1) = \begin{cases} \int_0^{x_1^2} \frac{3}{4} dx_2 = \frac{3}{4}x_1^2 & 0 \leq x_1 < 1 \\ \int_0^1 \frac{3}{4} dx_2 = \frac{3}{4} & 1 \leq x_1 \leq 2 \end{cases}$$

Expected value of X_1 :

$$\begin{aligned} E[X_1] &= \int_0^1 x_1 \left(\frac{3}{4} x_1^2 \right) dx_1 + \int_1^2 x_1 \left(\frac{3}{4} \right) dx_1 \\ &= \int_0^1 \frac{3}{4} x_1^3 dx_1 + \int_1^2 \frac{3}{4} x_1 dx_1 \\ &= \frac{3}{16} + \frac{9}{8} = \frac{21}{16} \end{aligned}$$

3. Expected value of $X_1 X_2$:

$$\begin{aligned} E[X_1 X_2] &= \int_0^1 \int_{\sqrt{x_2}}^2 \frac{3}{4} x_1 x_2 dx_1 dx_2 \\ &= \frac{3}{4} \int_0^1 x_2 \left(\int_{\sqrt{x_2}}^2 x_1 dx_1 \right) dx_2 \\ &= \frac{3}{8} \int_0^1 x_2 (4 - x_2) dx_2 = \frac{5}{8} \end{aligned}$$

4. Covariance:

$$\text{Cov}(X_1, X_2) = \frac{5}{8} - \left(\frac{9}{20} \right) \left(\frac{21}{16} \right) = \frac{5}{8} - \frac{189}{320} = \frac{11}{320}$$

Not surprisingly, these RVs are positively correlated.

Conditional Independence Example

Now, define $A = \{X_1 > 1\}$. Then $P(A) = 3/4$.

The conditional joint PDF is:

$$f_{X_1, X_2}(x_1, x_2 | A) = \begin{cases} \frac{f_{X_1, X_2}(x_1, x_2)}{P(A)} & (x_1, x_2) \in A \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1 & 1 < x_1 \leq 2, 0 \leq x_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Thus, the conditionals are independent uniform distributions: $f_{X_1}(x_1 | A) \sim U[1, 2]$ and $f_{X_2}(x_2 | A) \sim U[0, 1]$.

And we have:

$$f_{X_1, X_2}(x_1, x_2 | A) = f_{X_1}(x_1 | A) f_{X_2}(x_2 | A)$$

i.e., X_1 and X_2 are **conditionally independent** given A .

Consequently:

$$\text{Cov}(X_1, X_2 | A) = 0$$

which implies $E[X_1 X_2 | A] = E[X_1 | A] E[X_2 | A]$.